List of Abbreviations

Abbreviations of the model's species.

Short Notation	Full Name
GLUCIN	Intracellular glucose
GLUCOUT	Extracellular glucose - representative of the whole well
GLUT1	Glucose transporter~1
GLUT1_GLUCIN	GLUT1–GLUCIN complex
GLUT1_GLUCOUT	GLUT1–GLUCOUT complex
ATP	Adenosine triphosphate
ADP	Adenosine diphosphate
AMP	Adenosine monophosphate
AMPK	AMP-activated protein kinase
AMPKA	Active (phosphorylated) AMPK
AMPK_ATP	AMPK bound to ATP
AMPK_ADP	AMPK bound to ADP
AMPK_AMP	AMPK bound to AMP
AMPKA_ATP	Active AMPK bound to ATP
AMPKA_ADP	Active AMPK bound to ADP
AMPKA_AMP	Active AMPK bound to AMP
AMPK_t	Sum of inactive AMPK complexes
AMPKA_t	Sum of active AMPKA complexess
LKB1	Liver kinase~B1
PP2A	Protein phosphatase~2A
AMPK_LKB1	AMPK bound to LKB1
AMPK_ATP_LKB1	AMPK–ATP complex bound to LKB1
AMPK_ADP_LKB1	AMPK–ADP complex bound to LKB1
AMPK_AMP_LKB1	AMPK–AMP complex bound to LKB1
AMPKA_PP2A	Active AMPK bound to PP2A
AMPKA_ATP_PP2A	Active AMPK–ATP complex bound to PP2A
AMPKA_ADP_PP2A	Active AMPK–ADP complex bound to PP2A
AMPKA_AMP_PP2A	Active AMPK–AMP complex bound to PP2A
MTOR	Inactive mTORC1
MTORA	Active mTORC1
RHEB	Ras homolog enriched in brain
RHEBA	Active RHEB
TSC	Tuberous sclerosis complex
TSCA	Active TSC (GAP-active form)
ULK	Inactive ULK1/2
ULKA	Active ULK1/2
Q	Proton motive force

General Abbreviations of the Thesis.

Abbreviation	Full expression
AA	Amino acids
ACD	Accidental cell death
ATG	Autophagy-related gene/protein
AXP	Adenine nucleotide pool (ATP, ADP, AMP)
Bcl-2	B-cell lymphoma~2
Bcl-xL	B-cell lymphoma-extra large
BCRN	Biochemical reaction network
CMD	Correlation matrix distance
CRN	Chemical reaction network
cpc	Copies per cell
DMEM	Dulbecco's modified Eagle medium
EBSS	Earle's balanced salt solution
E	Enzyme (Michaelis–Menten)
S	Substrate (Michaelis–Menten)
ES	Enzyme–substrate complex (Michaelis–Menten)
P	Product (Michaelis–Menten)
GH	Growth hormone
IQR	Interquartile range
IP_3R	Inositol 1,4,5-trisphosphate receptor
KL	Kullback–Leibler divergence
K_M	Michaelis constant
LC3-I	Cytosolic form of LC3
LC3-II	Phosphatidylethanolamine-conjugated LC3
MIMO	Multiple-input multiple-output
MM	Michaelis–Menten
MOMP	Mitochondrial outer membrane permeabilization
NCI-60	National Cancer Institute 60 cell line panel
ODE	Ordinary differential equation
PCD	Programmed cell death
PE	Phosphatidylethanolamine
PI3K	Phosphoinositide 3-kinase
RV	Escoufier Rho-vector similarity
SSM	State-space model
UVRAG	UV radiation resistance-associated gene
V_{max}	Michaelis-Menten maximum enzymatic rate